WORKED SOLUTION

STEP III 1997, Question 3

Pure • Roots of Unity

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s3-q3

The key idea

The nth roots of unity $1, \omega, \omega^2, \ldots, \omega^{n-1}$ are exactly the roots of $z^n - 1$, so that polynomial factorises completely; dividing out the factor $z - 1$ gives the first identity. Put the polygon in the complex plane with O at the origin and the vertices become $re^{i\theta}$ times the roots of unity. Then the vector sum is a geometric series, and the squared chord lengths come from expanding $|\mathbf{a}_k - \mathbf{a}_1|^2$ with a dot product.

Solution

The product identity

For $k = 0, 1, \ldots, n-1$ we have $(\omega^k)^n = e^{2\pi i k} = 1$, so each ω^k is a root of $z^n - 1 = 0$. These n numbers are distinct, because their arguments $2\pi k/n$ are distinct and all lie in $[0, 2\pi)$. A polynomial of degree n with leading coefficient 1 and n distinct roots is the product of the corresponding linear factors, so

$$z^n - 1 = (z-1)(z-\omega)(z-\omega^2)\cdots(z-\omega^{n-1}).$$

On the other hand, summing a geometric series (or just multiplying out),

$$z^n - 1 = (z-1)(1 + z + z^2 + \cdots + z^{n-1}).$$

For every $z \neq 1$ we may divide both expressions by $z - 1$, which gives

$$(z-\omega)(z-\omega^2)\cdots(z-\omega^{n-1}) = 1 + z + z^2 + \cdots + z^{n-1} \qquad (z \neq 1).$$

Ask yourself The question says "any complex number z". What about $z = 1$, where we can't divide?

Both sides are polynomials of degree $n - 1$, and they agree at the infinitely many points $z \neq 1$. Their difference is then a polynomial with infinitely many roots, so it is the zero polynomial. (Alternatively, both sides are continuous in z, so let $z \to 1$.) Hence the identity holds for every complex z, including $z = 1$, as required.

The vector sum

Take the plane of the circle as the complex plane with O at the origin, and write a_k for the complex number representing A_k. If $a_1 = re^{i\theta}$, then labelling the vertices in order anticlockwise (consecutive vertices subtend angle $2\pi/n$ at O),

$$a_k = re^{i\theta}\,\omega^{k-1}, \qquad k = 1, 2, \ldots, n.$$

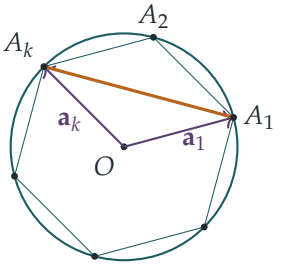
(Labelling clockwise instead just replaces ω by $\bar{\omega} = \omega^{n-1}$, which runs through the same set of roots; and the sum doesn't depend on the order of the vertices anyway.)

The vector $\overrightarrow{OA_k}$ corresponds to the complex number a_k, and adding vectors corresponds to adding complex numbers. So we need

$$\sum_{k=1}^{n} a_k = re^{i\theta}\left(1 + \omega + \omega^2 + \cdots + \omega^{n-1}\right).$$

The bracket is the right-hand side of the identity at $z = \omega$. Since ω appears among the factors on the left (for $n \geq 2$), the left-hand side has the factor $\omega - \omega = 0$, so the bracket is 0. (Equally, it is a geometric series with ratio $\omega \neq 1$, so it equals $(\omega^n - 1)/(\omega - 1) = 0$.) Therefore

$$\overrightarrow{OA_1} + \overrightarrow{OA_2} + \cdots + \overrightarrow{OA_n} = \mathbf{0}.$$



The sum of squared distances

Now write $\mathbf{a}_k = \overrightarrow{OA_k}$ as a vector, so $|\mathbf{a}_k| = r$ for every k and $\overrightarrow{A_1A_k} = \mathbf{a}_k - \mathbf{a}_1$. Expanding with the dot product,

$$\begin{aligned} |A_1A_k|^2 &= (\mathbf{a}_k - \mathbf{a}_1)\cdot(\mathbf{a}_k - \mathbf{a}_1) \\ &= |\mathbf{a}_k|^2 + |\mathbf{a}_1|^2 - 2\,\mathbf{a}_1\cdot\mathbf{a}_k \\ &= 2r^2 - 2\,\mathbf{a}_1\cdot\mathbf{a}_k. \end{aligned}$$

This also holds for $k = 1$, where both sides are 0. Summing over $k = 1, \ldots, n$ and using the fact that the dot product distributes over addition,

$$\begin{aligned} \sum_{k=1}^{n} |A_1A_k|^2 &= 2r^2n - 2\,\mathbf{a}_1\cdot\left(\sum_{k=1}^{n}\mathbf{a}_k\right) \\ &= 2r^2n - 2\,\mathbf{a}_1\cdot\mathbf{0} = 2r^2n, \end{aligned}$$

by the previous part. This is the required result.

Checks

- $n = 2$: the identity reads $z - (-1) = 1 + z$. The two points are ends of a diameter, so the sum is $0 + (2r)^2 = 4r^2 = 2r^2 \cdot 2$.
- $n = 4$, $r = 1$: the squared distances from one corner of the square are $0, 2, 4, 2$, totalling $8 = 2 \cdot 1^2 \cdot 4$.
- Numerically, for 2000 random $n \leq 20$, radii and rotations, the vector sum vanished and the sum of squared chords matched $2r^2 n$ to within 10^{-14}; the product identity was checked at random z (and at $z = 1$) to the same accuracy.

Common slips

- Dividing by $z - 1$ and forgetting that the identity is then only proved for $z \neq 1$.
- Writing $|\mathbf{a}_k - \mathbf{a}_1|^2 = |\mathbf{a}_k|^2 - |\mathbf{a}_1|^2$: the cross term $-2\,\mathbf{a}_1 \cdot \mathbf{a}_k$ is the whole point.

Beyond STEP

Putting $z = 1$ in the product identity and taking moduli gives $|1 - \omega|\,|1 - \omega^2| \cdots |1 - \omega^{n-1}| = n$: the product of the chord lengths from one vertex of a regular n-gon inscribed in a unit circle is exactly n, a neat companion to the sum of squares here. Factorising $z^n - 1$ over the roots of unity is the starting point for cyclotomic polynomials and for the discrete Fourier transform, both met early in a university course. Needham's *Visual Complex Analysis* is a beautiful place to see this kind of geometry done with complex numbers.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.